

**ISO/TC 12 "Quantities and units"**Secretariat: **SIS**Committee manager: **Cederholm Rebecca Ms****Draft Plenary Meeting Minutes**

Document type	Related content	Document date	Expected action
Meeting / Minutes	Meeting: VIRTUAL 26 May 2026	2026-06-09	INFO

Description

These draft minutes have been prepared following the meeting held on 26 May 2026 and are considered reviewed by the secretariat and Chair.

Members are invited to submit any comments or proposed corrections no later than **June 30th**.

If no comments are received by this deadline, the draft minutes will be considered approved.

ISO/TC 12, *Quantities and units*

Secretariat: SIS

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Report of the 23rd Meeting of ISO/TC 12 held on 2026-05-26 virtually via Zoom

1 Opening of the meeting (2026-05-26, 12:00 CEST)

2 Welcome from the host

Dr Torgny Carlsson, Chair of ISO/TC 12, welcomed everyone to the meeting.

3 Roll call of delegates

Delegates representing the following countries attended: *ASI (Austria), SAC (China), ICONTEC (Colombia), SFS (Finland), DIN (Germany), JISC (Japan), IPQ (Portugal), SIS (Sweden), BSI (United Kingdom)*

Liaisons: IUPAP, BIPM, CalConnect

17 delegates were present in total. Mr Olivier Pellegrino acted as Head of Delegation for IPQ (Portugal) on behalf of Mr Paulo Cabral.

See attendance list [N1628](#).

4 Work environment: [Presentation](#) on the ISO Code of Ethics and Conduct

Direct [link](#) to the ISO Code of Ethics and Conduct

5 Adoption of the agenda

The agenda was adopted as amended:

- It was noted that Mr. Sven Radhe was unable to attend and will be retiring in June. Ms. Rebecca Cederholm will replace him as Committee Manager. It was also noted that the Secretariat Support Team now comprises of Ms. Rebecca Cederholm, Mr. Jörgen Wyke and Ms. Anette Lindor Norén.
- Item 14: Ms. Mercè Ferrés Hernández (TPM), ISO/CS ([N1623](#)) sent her apologies, and Ms. Rebecca Cederholm presented in her place. Any questions pertaining to the presentation should be e-mailed to Ms. Rebecca Cederholm or Ms. Anette Lindor Norén.
- Item 18: A request was made by Mr. Olivier Pellegrino to add a liaison proposal to establish a liaison between ISO/TC 12 and ISO/IEC JTC 3, Quantum Technology. The item was added under "Any Other Business".

6 Appointment of the resolution drafting committee

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Dr Richard Brown, Ms Rebecca Cederholm, Mr Jörgen Wyke and Ms Anette Lindor Norén were appointed to the resolution drafting committee.

7 Report of the Committee Manager/Chair and the secretariat

Ms Rebecca Cederholm

See [N617](#): Report of the secretariat

Changes in membership: There have been changes in membership since the last ISO/TC 12 plenary meeting. Uzbekistan, O'ZTTSA has upgraded its status from an Observer (O) member to a Participating (P) member.

New liaison to be approved: IMU, International Mathematical Union was approved by ISO/TC 12 members and is currently awaiting approval by ISO Central Secretariat (ISO/CS). The liaison representative will be Prof Helge Holden.

8 Strategic issues / Review Strategic Business Plan vers.12

Ms. Rebecca Cederholm

Direct [link](#) to SBP of ISO/TC 12

The draft business plan ([N1609](#)) will be reformatted into ISO template and circulated via Committee Internal Ballot (CIB) for approval.

9 Working Groups:

IEC/TC 25/JWG 2 (Joint IEC/TC 25 and ISO/TC 12), Revision and amendment of IEC-related parts of ISO/IEC 80000-series

- SIS has decided to withdraw from its role as Secretariat for IEC/TC 25/JWG 2. Consequently, Prof. Göran Grimvall will step down as Convenor. To ensure continuity and a smooth transition, SIS will continue to support the work of JWG 2 and Prof. Grimvall until a new Secretariat and Convenor has been appointed. The Secretariat of IEC/TC 25 have been requested to initiate the process for appointing a new Secretariat and a new Convenor for JWG 2.

The work is progressing as planned, and no items required discussion at the time.

10 ISO/TC 12 participation in other activities

10.1 JCGM, Joint Committee for Guides in Metrology

Dr Carlsson, registered member in JCGM/WG GUM

See [N1613](#), SI JCGM/GUM Liaison report

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SI Digital Framework is developing machine-readable quantity expressions to improve digital interoperability. Synchronization with ISO 80000 recommended to ensure consistent representation of quantities and units and to support harmonized implementation across systems.

10.2 GUM, Guide to the Expression of Uncertainty in Measurement Prof. Leslie Pendrill

See [N1613](#), SI JCGM/GUM Liaison report

10.3 VIM, International Vocabulary of Metrology

VIM 4 is currently under voting as ISO/IEC Guide 99. It represents an update of the International Vocabulary of Metrology and is intended to strengthen the common conceptual framework for metrology by clarifying and modernizing key definitions.

11 Liaison reports

11.1 [CIE](#) International Commission on Illumination

Liaison representative : Dr. Torgny Carlsson

11.2 IEC/TC 1 Terminology

Liaison representative: Prof. Leslie Pendrill,

See [N1615](#), ISO/IEC JTC 1 Information technology /WG 15 Liaison report

Ongoing work on metrology (quality-assured measurement) in the IT domain is being addressed through ISO/IEC WG 2382 *Information technology – Vocabulary*:

- Part 1 (Fundamental terms) includes considerations on quantities and units in IT vocabulary and concept systems, physical properties of objects, IoT actuators, analogue/discrete systems, and analytic/clinical contexts.
- Part 4 (Data management) addresses units and terminology (“units and words”).
- Part 16 (Information theory) covers information and entropy.
- It is proposed to include “quantity” as an example property within the fundamental terms of Part 1.

A strong liaison is recommended to ensure alignment between the International System of Quantities and IT information modelling frameworks.

11.3 IEC/TC 25 Quantities and units

Liaison representative: Dr. Olivier Pellegrino

See [N1629](#), IEC/TC 25 Liaison report

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Joint Working Group 2 progressing on the IEC parts of the ISO/IEC 80000 series, with multiple parts currently in various stages of development. At this stage, no immediate decisions are required by TC12

11.4 [ISO/IEC JTC 1/SC 32](#) **Data management and interchange**

Liaison representative: Prof. Leslie Pendrill

11.5 [ISO/TC 37](#) **Language and terminology**

Liaison representative: Prof. Leslie Pendrill

11.6 [ISO/TC 43](#) **Acoustics**

Liaison représentative : Dr. Christopher J. Struck

See [N1608](#), ISO/ TC 43 Liaison Report

The 2025 systematic review of ISO 80000-8:2020 (Acoustics) identified no errors or needs for major additions. Subject to any issues raised by national committees, ISO TC 43 informally recommended affirmation.

IEC TC 29 (Electroacoustics) has published [a white paper](#) highlighting the magnitude of potential overestimation errors when uncertainty calculations are performed in decibels.

Both ISO TC 43 and IEC TC 29 note that the GUM does not provide worked examples for uncertainty evaluation of quantities expressed in levels, particularly regarding a standardized method for converting between level and fractional representations and back to final level values.

11.7 [ISO/TC 43/SC 1](#) **Noise**

Liaison représentative : Dr. Christopher J. Struck

See [N1608](#), ISO/ TC 43 Liaison Report

ISO TC 43/SC 1 (Noise) has approved a resolution to establish an ad hoc group to assess the need for revising Guide 2, *Treatment of Measurement Uncertainty*. This follows ongoing discussions on how uncertainty is handled in acoustic measurements expressed in logarithmic units (dB), where current practice in ISO TC 43/SC 1 standards differs from the GUM approach by calculating uncertainty directly in decibel space rather than first converting to fractional quantities.

11.8 [ISO/TC 43/SC 3](#) **Underwater acoustics**

Liaison representative: Secretariat

Contact details updated, and the secretariat is now held by DIN (Germany). The main areas of interest are ISO 80000-8 (Acoustics) and the development of the new Part 15.

11.9 [ISO/TC 69](#) **Applications of statistical methods**

Liaison representative: Secretariat

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The proposal for introducing rules on rounding numerical values was rejected, which would have resulted in the removal of Annex A from ISO 80000-1.

Interest remains in Part 2 (Mathematics), particularly with respect to statistical quantity designations. The next meeting scheduled for June 2026 in Berlin, Germany.

11.10 [ISO/TC 85](#) Nuclear energy, nuclear technologies, and radiological protection

Liaison representative: Secretariat

See [N1619](#), ISO/TC 85 Liaison report

ISO 80000-2 (Mathematics) and Part 10 (Atomic and nuclear physics) have been identified as relevant standards of interest.

The next meeting scheduled for August 2026 in Birmingham, USA

11.11 [ISO/TC 215](#) Health informatics

Liaison representative: Prof. Leslie Pendrill

See [N1614](#), ISO/TC 215 Liaison Report

The Health Informatics community remains strongly committed to UCUM (Unified Code for Units of Measure), with implementations deeply embedded in existing systems.

There is resistance to alternative digital unit systems, making comprehensive stakeholder consultation essential for any future changes.

A new three-year EXPERT project by LGC on pandemic diagnostic metrology is scheduled to start in summer 2026.

Other reports

Liaison report from OIML: [N1626](#)

The Health Informatics community remains strongly committed to UCUM (Unified Code for Units of Measure), with implementations widely embedded in existing systems. There is notable resistance to alternative digital unit systems, and any proposed changes will require thorough stakeholder consultation.

A new three-year EXPERT project led by LGC on pandemic diagnostic metrology is scheduled to start in summer 2026.

12 Review and confirmation of Liaisons Representatives

12.1 ISO/TC 85 Nuclear energy, nuclear technologies, and radiological protection, Draft resolution 8/2025: Proposal from China: Dr. Li Dehong as liaison representative

See [N1603](#), Request Liaison Representative in ISO/TC 85 and ISO/TC 69

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Dr Li introduced himself briefly to the committee.

Dr Li was confirmed as liaison to ISO TC85 (nuclear energy).

12.2 ISO/TC 69 Applications of statistical methods, Draft resolution 9/2025: Proposal from China: Dr Liu Zilong as liaison representative

See [N1603](#), Request Liaison Representative in ISO/TC 85 and ISO/TC 69

Dr Liu was confirmed as liaison to ISO TC69 (statistical methods)

13 Work program:

- 80000-2:2019, Quantities and units — Part 2: Mathematics

Call for Experts – Task Force for ISO 80000-2:2019 (Response Deadline: 30 April)
[N1592](#) and [N1606](#)

56 comments received from Germany and CalConnect.

An expert group has been formed, comprised by representatives from BIPM, BSI, SAC, ICONTEC, Japan, SIS, CalConnect, and potentially Standards Norway.

The project follows a 24-month development track, and preparation of the Working Draft (WD) is underway.

IEC Parts: Several parts of IEC 80000 are progressing through TC25 Joint Working Group 2. Status information is available on the ISO website.

A report was presented at the meeting by the Project Leader Mr Ronald Tse ISO/TC 12-smart ([N1624](#))

Systematic review results:

- ISO 80000-4: 2019 Quantities and units — Part 4: Mechanics

The committee reviewed the German comments ([N1599](#) and [N1597](#)). The comments will be considered and incorporated in the next revision.

Systematic review identified errors in Part 4 (mechanics); TC12 to launch CIB for revision, providing opportunity for expert nominations.

- ISO 80000-8:2020, Quantities and units — Part 8: Acoustics

The results of the systematic review were confirmed with no major changes required.

Result of Voting: Confirmed.

14 Update of ISO/IEC Directives regarding ISO/TC 12 rules

Ms Rebecca Cederholm held the presentation in lieu of Ms Mercè Ferrés Hernández (TPM), ISO/CS

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See N1623, *Changes to ISO/IEC Directives Part 1 2026*, and SMART

Important points to stress on the new DP1 at the Plenary:

- to consult TC 12 [N1616](#)
- to attend the webinars in July
- most impacting change is for the maintenance of Standards: shifting from the Systematic Reviews to the Stability Dates. The idea is that committees will take full ownership of the maintenance of the Standards. Applicable as of May 2027. Need to use the new SBP template.

Regarding SMART:

- TCs to work on OSD, as direct entry point to SMART.
- “smart” databases not yet available for working (e.g. for the revision of 80000-2)
- Information Model available via open license, feedback welcome

Other reports presented at the meeting from, Calconnect on ISO/TC 12-smart,
Mr. Ronald Tse

Smart Database Initiative: ISO/TC 12 to coordinate with ISO Central Secretariat to explore creating ISO/IEC 80000 as an information service ([N1624](#)).

The Chair thanked Mr Ronald Tse for his contribution.

15 Items of future work

Nothing to report

16 Communication

Nothing to report

17 Next ISO/TC 12-meeting 2026

Virtual plenary is scheduled for May 25, 2027, starting at 1300 CEST to accommodate global participation.

18 Any other business

Proposal to establish a liaison between ISO/TC 12 and ISO/IEC JTC 3 Quantum Technology, *Mr. Olivier Pellegrino*

CIB to be created for the request of liaison from TC 12 to ISO/IEC JTC 3.

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19 Approval of resolutions

Drafting Committee: Ms Rebecca Cederholm (Committee Manager), Mr Jörgen Wyke (Secretariat support) and Ms Anette Lindor Norén (Secretariat support). *Dr Brown left the meeting prior to the drafting phase and was therefore unable to participate in the drafting committee.*

Draft resolutions 2–10/2026 were approved unanimously; see [N1627](#).

20 Closure of the meeting

The Chair thanked all delegates and closed the meeting at 15:07 CEST.